FINTECH ARCHITECTURE | ACADEMIC & INDUSTRY DEFENSE | 4-MINUTE (240S) RUNTIME | PRODUCTION DEMO READY

TRUSTROUTE – Intelligent Fraud-Aware Payment Routing System

Complete 4-Minute Demonstration Video Script, Scene-by-Scene Storyboard & Technical Production Guide

Target Runtime: 4m 00s (240 Seconds Exactly)

Target Audience: Faculty, Project Evaluators, Engineers

Technology Stack: Next.js 16, Node/Express, MongoDB, Neo4j

Core Algorithm: Yen's K-Shortest Paths (KSP) + CAMARA

"Trust Every Transaction. Route Every Payment Intelligently."

A proposed unified platform combining real-time transaction risk scoring, telecom identity telemetry, and graph-based multi-criteria payment network optimization.

1. Project Identity & Technical Foundation

Core Objective: Traditional financial payment routing focuses almost exclusively on finding a reachable pathway between banking endpoints. However, it fails to dynamically integrate real-time transaction risk, telecom-level identity indicators, and live network degradation into routing decisions. **TrustRoute** bridges this gap by unifying transaction risk assessment, telecom identity signals (CAMARA Open Gateway), and graph-based multi-path alternative calculation (Yen's K-Shortest Paths algorithm) into an integrated enterprise solution.

> **Key Academic & Technical Honesty Note:** TrustRoute is an *integrated proposed solution*. CAMARA telecom APIs are operated via an enterprise-grade mock service demonstrating SIM Swap, Device Swap, and Location Verification. Yen's algorithm calculates true loopless alternative routes across a shared Neo4j topology. All transaction, user, and audit data are persisted in live MongoDB instances.

2. System Architecture & Technology Stack

Component	Technology Used	Engineering Purpose & Implementation Details
Frontend Interface	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS	Role-scoped dashboards for Corporate Senders, Fraud Analysts, and Infrastructure Admins. Zero-PII masking, real-time telemetry, and graph visualization.
Backend API Gateway	Node.js, Express, TypeScript, JWT, RBAC	Restful transaction orchestration, multi-tiered permission middleware, centralized audit logging, and payment provider dispatch.
Transactional Database	MongoDB & Mongoose ODM	Stores users, transaction lifecycle states, risk assessment records, review queues, audit trails, and revoked session tokens.
Network Graph Topology	Neo4j Graph Database (Cypher)	Models interconnected banks and payment gateways as graph nodes; directed rails as edges with cost, latency, risk, and active status.

Component	Technology Used	Engineering Purpose & Implementation Details
Identity & Telecom Engine	CAMARA Open Gateway (Mock Implementation)	Simulates telecommunication security checks: SIM Swap (recent IMSI changes), Device Swap (IMEI drift), and Geo-location radius verification.
Routing Engine	Dijkstra & Yen's K-Shortest Paths (KSP)	Calculates up to *K* loopless alternative payment pathways ranked by composite weight: `W = w1*Cost + w2*Latency + w3*Risk`.
Decisioning Engine	Rule-based Multi-Factor Risk Classifier	Produces deterministic outcomes: **APPROVED** (eligible to route), **STEP-UP** (requires MFA/review), or **BLOCKED** (hard stop).

3. Master 4-Minute (240 Seconds) Timeline Overview

Timestamp	Duration	Scene Title	Module & Route	Primary Evaluator Focus
0:00 – 0:25	25 sec	Scene 1: Introduction & Identity	Landing Page / Brand Splash	Clear problem statement & brand authority
0:25 – 0:55	30 sec	Scene 2: Problem & Proposed Solution	Architectural Flow & Tri-State Engine	Why risk must precede payment routing
0:55 – 1:20	25 sec	Scene 3: System Architecture & Tech	Full-Stack Architecture Overview	Decoupled React/Express/MongoDB/Neo4j stack
1:20 – 1:55	35 sec	Scene 4: Sender Module Walkthrough	`/payment` & `/transactions`	Initiating transfer, CAMARA checks, decision
1:55 – 2:35	40 sec	Scene 5: Analyst Module & Escalation	`/analyst`, `/risk`, `/review-queue`	Risk intelligence, contributing factors, escalate
2:35 – 3:20	45 sec	Scene 6: Admin Module & Infrastructure	`/admin`, `/admin/users`, `/network`	Network topology, zero-PII users, governance
3:20 – 3:45	25 sec	Scene 7: Routing & Yen's Algorithm	`/admin/simulation` (KSP Engine)	K-shortest paths ranking, multi-hop routes
3:45 – 4:00	15 sec	Scene 8: Conclusion & Final Motto	TrustRoute Closing Screen	Memorable punchline and impact summary
TOTAL: 240 Seconds		**100% Calibrated 4-Minute Technical Demonstration**		

4. Scene-by-Scene Screenplay & Narration Guide

SCENE 1: Introduction, Problem Hook & Project Identity

`0:00 – 0:25 (25 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Display:** High-resolution TrustRoute logo, project title, and motto: "*Trust Every Transaction. Route Every Payment Intelligently.*"
- **Action:** Smooth fade-in to the TrustRoute application home dashboard. Show subtle cursor movement across clean navigation bar.
- **Text Overlay:** "*TrustRoute: Intelligent Fraud-Aware Payment Routing System*"
- **Camera Movement:** Static wide shot, gentle 1.05x zoom toward the central system health badge.

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"*In modern digital finance, moving money is fast—but verifying whether a transaction is legitimate, and finding the most resilient rail to send it across, remain disconnected challenges.*

Welcome to TrustRoute—an intelligent, fraud-aware payment routing platform. Our mission: 'Trust Every Transaction. Route Every Payment Intelligently.'

TrustRoute unites real-time identity signals, risk scoring, and graph-based network optimization to ensure secure, uninterrupted payments."

Evaluator Takeaway: Establishes immediate professional fintech credibility; introduces the core innovation of coupling risk assessment directly to payment pathway calculation.

SCENE 2: The Core Challenge & Proposed Tri-State Solution

`0:25 – 0:55 (30 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Display:** Clean visual flow diagram or interactive UI showing the 5-stage pipeline: `Transaction Initiated → Risk Assessment → Routing Decision → Intelligent Pathway → Result`.
- **Action:** Highlight the 3 deterministic outcomes produced by the Risk Engine:
  • **APPROVED:** Eligible for payment routing.
  • **STEP-UP:** Elevated risk; biometric/telecom challenge required.
  • **BLOCKED:** High fraud probability; immediate terminal halt.
- **Text Overlay:** "*Tri-State Risk Decisioning Engine*"

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"*Traditional payment gateways blindly route funds based on static agreements—leaving institutions vulnerable to SIM swapping, account takeovers, and downstream gateway congestion.*

TrustRoute introduces a proactive paradigm: we evaluate risk before a single penny moves.

Every transaction is scrutinized through telecom identity signals and behavioural parameters, yielding one of three precise decisions: Approved for routing, Step-Up verification required, or immediately Blocked. Only verified transactions enter our intelligent routing pipeline."

Key Concept: Emphasizes that route finding and fraud evaluation are decoupled in logic but coupled in workflow—risk gates routing.

SCENE 3: System Architecture & Technology Stack

`0:55 – 1:20 (25 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Display:** High-level system architecture graphic or quick pan across the modular interface.
- **Highlighting Elements:**
  • *Frontend:* Next.js 16 App Router & Tailwind CSS.
  • *Backend:* Express REST API with TypeScript & JWT RBAC.
  • *Database:* MongoDB for users, transactions, and audit logs.
  • *Graph Engine:* Neo4j modeling dynamic banking topology.
  • *Routing Core:* Yen's K-Shortest Paths algorithm.
  • *Identity:* CAMARA Open Gateway mock integration.

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"Under the hood, TrustRoute is engineered as a modern, decoupled enterprise architecture.

The frontend is built with Next.js 16 and TypeScript for responsive, role-scoped operations. Backend orchestration is handled by an Express and Node.js microservice architecture.

MongoDB persists transaction lifecycles and comprehensive audit trails, while a high-performance Neo4j graph database models the interconnected payment network.

Integrated with CAMARA telecom APIs and Yen's K-Shortest Paths algorithm, TrustRoute solves routing with mathematical precision."

Evaluator Takeaway: Demonstrates real engineering breadth: relational document storage paired with Cypher-based graph analytics.

SCENE 4: Corporate Sender Module Demonstration

`1:20 – 1:55 (35 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Navigation:** Open browser at `http://localhost:3000/` (logged in as Sarah Jenkins, Corporate Sender).
- **Action 1:** Show Sender Dashboard: Available Balance ($142,500), transfer metrics, and clean top navigation.
- **Action 2:** Click **New Transfer** in sidebar (navigates to `/payment`).
- **Action 3:** Fill transfer form: Source Bank: *JPMorgan Chase*, Destination: *Barclays UK*, Amount: *$25,000 USD*.
- **Action 4:** Click **"Submit Transfer"**. Show live risk evaluation dialog: SIM Swap check: Valid, Device Check: Match. Outcome: **APPROVED**.
- **Action 5:** Transition to `/transactions` showing updated ledger status.

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"Let's look at the live platform. In the Sender Portal, corporate users manage liquidity and initiate transfers with complete transparency.

When a user initiates a twenty-five-thousand-dollar transfer from JPMorgan to Barclays, TrustRoute immediately triggers pre-routing telemetry.

In milliseconds, our CAMARA service verifies device integrity and recent SIM activity. The transaction passes the risk threshold and is marked APPROVED.

The sender receives immediate confirmation, with the transaction seamlessly recorded on their ledger without operational friction."

Demo Safety: Utilizes seeded, realistic mock corporate entities without exposing real personally identifiable information.

SCENE 5: Fraud Analyst Module & Escalation Workflow

`1:55 – 2:35 (40 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Navigation:** Switch persona or log in as Alex Rivera, Fraud Analyst (navigates to `/analyst`).
- **Action 1:** Highlight Analyst Dashboard: Total Flags, High-Risk Queue, Anomaly Velocity.
- **Action 2:** Click **Risk Intelligence** (`/risk`). Open transaction detail drawer for a flagged transfer.
- **Action 3:** Show *Contributing Risk Factors*: SIM swapped within 24 hours (Weight 40%), Geolocation deviation > 500km (Weight 35%). Composite Risk Score: **88/100**.
- **Action 4:** Click **"Transfer to Admin"** button. Show instantaneous modal confirmation: "*Case Escalated to Admin Review Queue*".

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"Now, let's step into the shoes of a Fraud Analyst.

In the Analyst Console, suspicious activities are flagged instantly. Opening Risk Intelligence reveals a transaction flagged with a risk score of 88 out of 100.

Rather than relying on black-box predictions, TrustRoute provides clear explainability: a recent SIM swap within 24 hours combined with a drastic geolocation mismatch.

Recognizing potential fraud, the analyst clicks 'Transfer to Admin'—seamlessly escalating the full forensic package into the administrative investigation pipeline."

Feature Highlight: Demonstrates cross-role collaboration between tier-1 fraud analysts and senior system administrators via backend database persistence.

SCENE 6: Administrator Console, Network Topology & User Oversight

`2:35 – 3:20 (45 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Navigation:** Switch to Elena Rostova, Administrator (navigates to `/admin`).
- **Action 1:** Show Admin Dashboard with system-wide KPI metrics and active rails.
- **Action 2:** Click **Review Queue** (`/review-queue`). Show the escalated case received from the Analyst; demonstrate the dropdown status selector (Pending → Under Review → Blocked).
- **Action 3:** Click **User Management** (`/admin/users`). Point out Zero-PII compliance: masked emails, hashed phone numbers, and role controls.
- **Action 4:** Click **Graph / Topology** (`/network`). Showcase the interactive Neo4j visualizer showing banks, payment gateways, and active interconnecting edges.

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"As an Administrator, TrustRoute provides full operational oversight.

In the Review Queue, the case escalated by our analyst appears immediately. Administrators can filter by status, review forensic signals, and enforce terminal resolutions like 'Blocked' or 'Approved'.

In User Management, enterprise privacy is built-in: sensitive customer data is strictly anonymized with zero PII exposure, displaying only masked IDs and volume metrics.

Finally, in Network Topology, we see the real-time payment ecosystem rendered from Neo4j—showing banks, intermediary gateways, latency benchmarks, and active routing corridors."

Governance Highlight: Shows end-to-end auditability, regulatory privacy compliance (GDPR/Zero-PII), and live graph data synchronization.

SCENE 7: Intelligent Payment Routing & Yen's Algorithm Highlight `3:20 – 3:45 (25 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Navigation:** Navigate to **Simulator** (`/admin/simulation`).
- **Action 1:** Select Source: *Deutsche Bank*, Destination: *HSBC UK*. Amount: *50,000 USD*. Set **K-Value = 3**.
- **Action 2:** Click **"Calculate Pathways"**.
- **Action 3:** Camera zoom onto the calculated results table:
  • *Rank 1 (Primary):* Deutsche Bank → SWIFT Direct → HSBC UK (Cost: $12.50, Latency: 180ms)
  • *Rank 2 (Alternative 1):* Deutsche Bank → SEPA Instant → ClearingHouse → HSBC UK (Cost: $16.00, Latency: 240ms)
  • *Rank 3 (Alternative 2):* Deutsche Bank → Fedwire Relay → HSBC UK (Cost: $21.00, Latency: 310ms)

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"The technical crown jewel of TrustRoute is our intelligent routing engine powered by Yen's K-Shortest Paths algorithm.

When we simulate a fifty-thousand-dollar transfer with a K-value of three, TrustRoute doesn't just return a single path—it computes the top three distinct, loopless pathways across the Neo4j graph.

Each path is ranked against a composite cost, latency, and risk metric. If primary rail congestion strikes, payment orchestrators have mathematically verified backup routes instantly ready."

Algorithm Distinction: Dijkstra finds only a single path; Yen's KSP provides resilient alternative failover routes ranked by multi-objective weights.

SCENE 8: Conclusion & Memorable Value Proposition `3:45 – 4:00 (15 Seconds)`

VISUAL DISPLAY & SCREEN ACTION

- **Display:** Return to the clean branded closing graphic featuring TrustRoute logo and technology badges.
- **Bullet Fade-in:**
  • Pre-Routing Risk Assessment
  • CAMARA Telecom Identity Telemetry
  • Neo4j Graph Optimization
  • Yen's K-Shortest Path Resilience
- **Final Screen:** TrustRoute Motto in bold glowing typography.

VOICE-OVER NARRATION (NATURAL, AUTHORITATIVE)

"By uniting transaction intelligence with payment network graph optimization, TrustRoute redefines modern payment infrastructure.

TrustRoute brings transaction intelligence and payment network intelligence together—helping make every payment decision smarter, resilient, and more secure.

Thank you."

Ending Impact: Concludes exactly on the 240-second mark with the project motto, creating a lasting professional impression.

5. Evaluator Defense: Technical & Algorithmic Deep Dive

How Yen's K-Shortest Paths Algorithm Works in TrustRoute

In standard payment networks, simple Dijkstra or A* algorithms only produce the single cheapest or fastest route. If that rail fails or suffers latency spikes, transactions stall. TrustRoute implements **Yen's K-Shortest Paths Algorithm** on top of Neo4j to compute *K* loopless alternative pathways:

Step 1 (Root Shortest Path): Uses Dijkstra's algorithm to determine the global shortest path A_1 between source bank S and destination bank D using composite edge weights $W(e) = \alpha \cdot \text{Cost} +$

\beta \cdot \text{Latency} + \gamma \cdot \text{Risk} \).
Step 2 (Spur Node Iteration): For finding the \(k \)-th path (\(2 \le k \le K \)), the algorithm iterates through each node in path \(A_{k-1} \) as a *spur node*. Edge costs previously used are temporarily masked to prevent duplicate sub-paths.
Step 3 (Candidate Heap): A spur path is computed from the spur node to the destination, concatenated with the root path, and stored in a candidate priority queue \(B \).
Step 4 (Loopless Selection): The lowest-cost loopless path from \(B \) is extracted as \(A_k \). This guarantees mathematically distinct alternative routes ready for failover.

6. Pre-Recording & Production Verification Checklist

Category	Verification Task	Status
Environment Setup	Backend running on `localhost:5000`; Frontend running on `localhost:3000`	Verified [OK]
Database Seeding	MongoDB seeded with test transactions, users, and audit logs (`npm run seed`)	Verified [OK]
Graph Engine	Neo4j connected and populated with 10+ bank nodes and interconnecting edges	Verified [OK]
Screen Resolution	Recording display locked to 1920x1080 (1080p), 60 FPS, browser zoom at 100%	Standard [Ready]
Audio Calibration	High-quality USB condenser mic, noise gate applied, target cadence ~130 words/min	Calibrated [Ready]
Timing Rehearsal	Each scene checked against its exact allocated seconds for a strict 240s runtime	Timed [240s Total]